

In the following exercises, you will be asked to determine whether a given stochastic process is a martingale, a submartingale, or a supermartingale. The martingale property is a special case of the more general submartingale and supermartingale properties.

Suppose $\{X_n\}_{n \geq 0}$ is a stochastic process adapted to a filtration $\{\mathcal{F}_n\}_{n \geq 0}$. Recall that X_n is a martingale if $E[X_{n+1} | \mathcal{F}_n] = X_n$ for all $n \geq 0$. Similarly, X_n is a submartingale if $E[X_{n+1} | \mathcal{F}_n] \geq X_n$ and a supermartingale if $E[X_{n+1} | \mathcal{F}_n] \leq X_n$. For each of the following processes, determine whether it is a martingale, submartingale, or supermartingale. If it is, state the martingale property that holds. If it is not, state the property that fails.

1. Let $\{X_n\}_{n \geq 0}$ be a stochastic process adapted to a filtration $\{\mathcal{F}_n\}_{n \geq 0}$ such that $X_0 = 1$ and $X_n = 2^{S_n}$ for all $n \geq 0$, where $S_n = \sum_{i=0}^{n-1} X_i$. Determine whether $\{X_n\}_{n \geq 0}$ is a martingale, submartingale, or supermartingale. If it is, state the martingale property that holds. If it is not, state the property that fails.

2. Let $\{X_n\}_{n \geq 0}$ be a stochastic process adapted to a filtration $\{\mathcal{F}_n\}_{n \geq 0}$ such that $X_0 = 1$ and $X_n = \prod_{i=0}^{n-1} (1 + X_i)$ for all $n \geq 0$. Determine whether $\{X_n\}_{n \geq 0}$ is a martingale, submartingale, or supermartingale. If it is, state the martingale property that holds. If it is not, state the property that fails.

3. Let $\{X_n\}_{n \geq 0}$ be a stochastic process adapted to a filtration $\{\mathcal{F}_n\}_{n \geq 0}$ such that $X_0 = 1$ and $X_n = \prod_{i=0}^{n-1} (1 - X_i)$ for all $n \geq 0$. Determine whether $\{X_n\}_{n \geq 0}$ is a martingale, submartingale, or supermartingale. If it is, state the martingale property that holds. If it is not, state the property that fails.

4. Let $\{X_n\}_{n \geq 0}$ be a stochastic process adapted to a filtration $\{\mathcal{F}_n\}_{n \geq 0}$ such that $X_0 = 1$ and $X_n = \prod_{i=0}^{n-1} (1 + X_i^2)$ for all $n \geq 0$. Determine whether $\{X_n\}_{n \geq 0}$ is a martingale, submartingale, or supermartingale. If it is, state the martingale property that holds. If it is not, state the property that fails.

16.4 Stochastic Processes with Independent Increments

Let $\{X_n\}_{n \geq 0}$ be a stochastic process with independent increments. Recall that this means that for any $n \geq 0$ and any $k \geq 1$,

$$X_{n+k} - X_n = (X_{n+1} - X_n) + (X_{n+2} - X_{n+1}) + \dots + (X_{n+k} - X_{n+k-1})$$

and the increments $X_{n+1} - X_n, X_{n+2} - X_{n+1}, \dots, X_{n+k} - X_{n+k-1}$ are independent. Similarly, the increments $X_1 - X_0, X_2 - X_1, \dots, X_k - X_{k-1}$ are independent. For each of the following processes, determine whether it has independent increments. If it does, state the increments that are independent.

1.

$$X_n = \sum_{i=0}^{n-1} Z_i^2, \quad Z_i \sim N(0, 1)$$

2. Let $\{X_n\}_{n \geq 0}$ be a stochastic process with independent increments. Recall that this means that for any $n \geq 0$ and any $k \geq 1$,